



COMPUTER

- All digital circuits can be realized by using only
 - Exclusive OR gates
 - Halfadders
 - Multiplexers
 - OR gate
- The Boolean function $a + (a \cdot b)$ is equivalent to
 - $a \cdot b$
 - $a + b$
 - $a \cdot \bar{b}$
 - $\bar{a} + b$
- Which of the following circuit is used as a memory device in computers?
 - Flip-Flop
 - Rectifier
 - Comparator
 - All of these
- Convert the hexadecimal number 4DF to its octal equivalent.
 - 2333
 - 2337
 - 2773
 - 2373
- A tautology is a Boolean formula that is always true. What of the following is a tautology?
 - x
 - $(\bar{x} + \bar{x})y$
 - $x + \bar{y} + \bar{x}$
 - $(xy) + \bar{x}$
- Acronym of EEPROM is
 - Extended Erasable Programmable Memory
 - Electrically Erasable Read Only Memory
 - Electrically Erasable Programmable Read Only Memory
 - Extended Erasable Page-oriented Memory
- For reproducing sound, CD audio player uses a
 - Quartz crystal
 - Titanium
 - Barium ceramic
 - Laser beam
- When we open an internet site, we see WWW. What does WWW Stand for ?
 - World Wide Word
 - World Wide Web
 - World Wide Webinar
 - Word Widing Works
- The answer of the operation $(10111)_2 \times (1110)_2$ in hex equivalent is
 - 150
 - 14 C
 - 142
 - 13 E
- The minimum number of bits to represent character from ASCII code set is
 - 2
 - 8
 - 5
 - 7

ANSWER

1.	B	2.	B	3.	A	4.	B	5.	C
6.	C	7.	B	8.	B	9.	C	10.	D

SOLUTION

- (b)
All digital circuits can be realized by using only half adders. A half adder consists of an XOR gate and an AND gate.

2. (b)

The Boolean function $a + (a' \cdot b)$ is solved as
 $a + (a' \cdot b)$
 $= (a' + a) \cdot (a + b)$ [using distributive law]
 $= 1 \cdot (a + b)$ [$\because a + a' = 1$]
 $= a + b$

3. (a)

The flip-flop circuit is used as a memory device in computers.

4. (b)

Conversion of hexadecimal 4DF to its octal as

4	D	F	
0100	1101	1111	$= (010011011111)_{16}$

Now, $(010011010000)_{16}$ into octal number is

$$\frac{010}{2} \frac{011}{3} \frac{011}{3} \frac{111}{7} = (2337)_8$$

5. (c)

As we know a tautology is a Boolean formula that is always true. From the giving options we solve all through the truth table. Here $x + \bar{y} + \bar{x}$ is satisfying the condition of tautology.

x	y	$\bar{x}$	$\bar{y}$	$x + \bar{y} + \bar{x}$
0	0	1	1	1
0	1	1	0	1
1	0	0	1	1
1	1	0	0	1

The output of $x + \bar{y} + \bar{x}$ is true.

6. (c)

EEPROM stands for Electrically Erasable programmable Read only Memory. It is a type of a non-volatile memory used in computers and other electronic devices.

7. (d)

Compact Disc (CD) is a disc which contains digital data that is scanned by a laser beam for the reproduction of recorded sound.

8. (b)

WWW stands for World Wide Web.

9. (c)

The multiplication of $(10111)_2 \times (1110)_2$ is
 $(10111)_2 \times (1110)_2 = (000101000010)_2$

Now, the result is converted into hexadecimal as

$$\frac{0001}{1} \frac{0100}{4} \frac{0010}{2}$$

So, the answer of the operation

$(10111)_2 \times (1110)_2$ in hex equivalent is $(142)_{16}$.

10. (d)

The minimum number of bits to represent a character from ASCII code set is 7.

